

Trends in Cognitive Function Before and After Diabetes Onset

The China Health and Retirement Longitudinal Study

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Abstract

Background and Objectives

Individuals with prevalent diabetes were known to have a higher risk of dementia and lower cognitive function. However, trends of cognitive function before diabetes and in the short term after new-onset diabetes remain unclear.

Methods

This study included participants without baseline diabetes from the China Health and Retirement Longitudinal Study. Cognitive tests were conducted at baseline (wave 1) and at least one time from wave 2 (2013) to wave 4 (2018). Cognitive function was assessed using a global cognition score which was the summary measure of 4 cognitive tests. A linear mixed model was constructed to fit the trends in cognitive function before and after diabetes onset and the trends among nondiabetes. The threshold of statistical significance was $p < 0.05$.

Results

During the 7-year follow-up, 1,207 (9.7% of 12,422, 59.1 ± 8.6 years, 39.9% male participants) participants developed new-onset diabetes. The cognitive function of both the without diabetes group and the diabetes group declined annually during the follow-up. The annual decline rate of the diabetes group before diabetes onset was similar to that of the without diabetes group during the whole follow-up period. After diabetes onset, participants experienced statistically significant faster cognitive declines in global cognition (-0.023 SD/year; 95% CI -0.043 to -0.004 ; $p = 0.019$) and visuospatial abilities test (-0.036 SD/year; -0.061 to -0.011 ; $p = 0.004$), but not in tests of episodic memory (-0.018 SD/year; -0.041 to 0.004 ; $p = 0.116$), attention and calculation (-0.017 SD/year; -0.037 to 0.003 ; $p = 0.090$), or orientation (0.001 SD/year; -0.018 to 0.020 ; $p = 0.894$), compared with the cognitive slope before diabetes. In subgroup analysis, compared with those who developed diabetes between 45–54 years, those developing diabetes older (55–64 years, p for interaction = 0.701; 65–74 years, p for interaction = 0.996) did not demonstrate different rates of global cognitive decline after diabetes.

Discussion

Individuals experienced faster rate of cognitive decline in a few years after diabetes onset, but not during the prediabetes period. Age did not modify the effect of diabetes on postdiabetes cognitive decline. Efforts in eliminating the adverse impacts on cognition should be started on diagnosis of diabetes.

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Glossary

ApoE = apolipoprotein E; **BMI** = body mass index; **CHARLS** = China Health and Retirement Longitudinal Study; **CHAP** = Chicago Health and Aging Project; **ELSA** = English Longitudinal Study of Ageing; **HbA1c** = hemoglobin A1c; **IADLs** = instrumental activities of daily living; **MCI** = mild cognitive impairment; **MMSE** = Mini-Mental State Examination; **NFTs** = neurofibrillary tangles; **NPs** = neurotic plaques.

Introduction

Dementia currently affects approximately 50 million people worldwide. This number is predicted to increase to 152 million by 2050.¹ Owing to the lack of effective treatments, modifiable risk factors have drawn special attention for dementia prevention. Diabetes is a key modifiable risk factor for dementia.² Hyperglycemia contributes to dementia through both Alzheimer disease–related and vascular disease–related pathology.^{3,4}

Diabetes increased the risk of follow-up dementia.⁵ Meanwhile, those with known diabetes had lower cognitive function cross-sectionally and faster cognitive decline longitudinally.^{6,7} Most studies only interviewed diabetes status at baseline, reflecting the cognitive trends among participants with prevalent diabetes. However, hyperglycemia might develop several years before the cognitive assessment. Cognitive function before or in the short period after diabetes onset is unknown. Considering that prediabetes could also affect cognitive health, cognitive deterioration might begin even before diabetes.⁸ Dementia is a syndrome characterized by declines in cognitive function. Understanding the cognitive trajectories before and in the short period after new-onset diabetes could help unravel the pathophysiology of diabetes on the brain and the prevention and management of dementia in diabetic individuals.

Few studies have indeed reported cognitive trends after^{9–13} or before and after new-onset diabetes.^{14,15} However, several knowledge gaps remain unaddressed. First, the study population was relatively older in the previous 2 studies which calculated the cognitive trends before and after diabetes.^{14,15} A recent article published in the JAMA reported younger age at onset of diabetes was even associated with higher risk of subsequent dementia.¹⁶ Therefore, whether individuals with lower mean age exhibited different magnitude of cognitive decline is not clear. Second, to our knowledge, cognitive trajectories before and after diabetes onset have not been investigated in the Chinese population, who have different educational and economic levels. This might contribute to different cognitive reserve and thus different patterns of cognitive change.⁸ Third, previous studies achieved mixed results in test-specific cognitive trends. For example, some reported accelerated declines in memory function after diabetes onset,^{14,15} while some reported that incident diabetes was not associated with longitudinal memory decline.^{10,11}

In this study, we aimed to explore the temporal pattern of declines in cognitive function before and after diabetes onset

among middle-aged and older Chinese population (45 years or older).

Methods

Study Participants

This study used data from wave 1 to wave 4 of the China Health and Retirement Longitudinal Study (CHARLS), a nationally representative and community-based cohort study in China.¹⁷ At baseline (defined as wave 1), 17,708 participants were recruited from 150 national units distributed in 28 provinces of China. The wave 1 survey was performed between June 2011 and March 2012. Three follow-up visits (wave 2, 2013; wave 3, 2015; and wave 4, 2018) were conducted after wave 1.

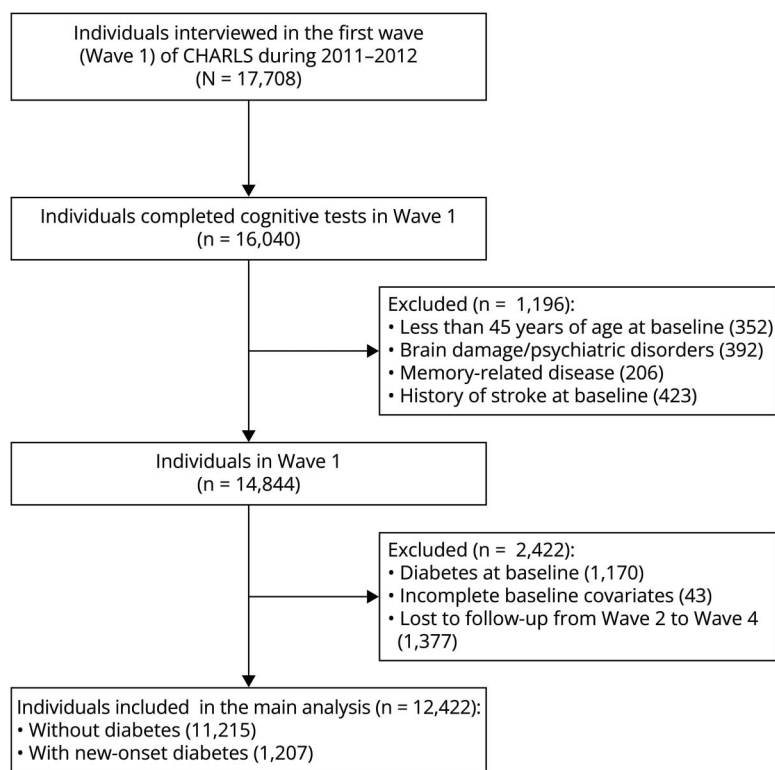
The flowchart for sample selection is shown in Figure 1. Of the 16,040 participants who finished cognitive assessments at baseline, 1,196 were excluded for the following reasons: under 45 years at baseline and having a history of brain injuries or psychiatric disorders, stroke, or memory-related diseases which included Alzheimer disease, brain atrophy, and Parkinson disease. Subsequently, 2,422 of the remaining 14,844 individuals were further excluded due to diabetes history at baseline ($n = 1,170$), lack of baseline covariates ($n = 43$), or not receiving at least one cognitive assessment during the follow-up from wave 2 to wave 4 ($n = 1,377$).

Standard Protocol Approvals, Registrations, and Patient Consents

CHARLS was approved by the Ethics Review Committee of Peking University (IRB00001052–11014 and IRB00001052–11015). Each participant had signed 2 copies of informed consent (eAppendix 1, [links.lww.com/WNL/D458](https://www.lww.com/WNL/D458)).

Assessment of Cognitive Function

Cognitive function was evaluated in each wave and included the following 4 tests: episodic memory, visuospatial abilities, orientation, and attention and calculation. The episodic memory was tested by word recall, and its total score was composed of the correct number of immediate and delayed (5 minutes) word recalls. The score ranged from 0 to 10 (0.5 points for each correct word). Visuospatial abilities were evaluated by replicating 2 overlapping pentagons in a drawing. Participants who completed the task successfully received 1 point. The orientation abilities were assessed by asking participants to identify the date (year, month, day), day of the week, and season (score ranging, 0–5). The attention and

Figure 1 Flowchart of Sample Selection

calculation abilities were measured by repeatedly subtracting 7 from 100 for 5 times (score ranging, 0–5). The reliability and validity of these tests have been well documented in the previous investigations.^{17,18} Z-scores were estimated by the mean and SD of the baseline scores, allowing for direct comparisons across different cognitive tests. Hence, a Z-score of 1 would describe cognitive performance that is 1 SD above the mean score at the baseline. We generate a global cognitive Z-score by summing the Z-scores of the above 4 tests and restandardizing it to baseline in the same manner.¹⁹

Assessment of Diabetes Onset

Diabetes was defined as self-reported physician's diagnosis, antidiabetic medication use, or hemoglobin A_{1c} (HbA_{1c}) $\geq 6.5\%$. Information on self-reported diagnosis and medication use was available in all 4 waves. The HbA_{1c} level was measured in wave 1 and wave 3. For some participants, the time of diabetes onset was recorded through interviews ("When was diabetes first diagnosed or known?"). For other participants, the time of diabetes onset was recorded as the midpoint between the last time without diabetes and the first time with diabetes. The study did not distinguish between type 1 and type 2 diabetes.

Covariates

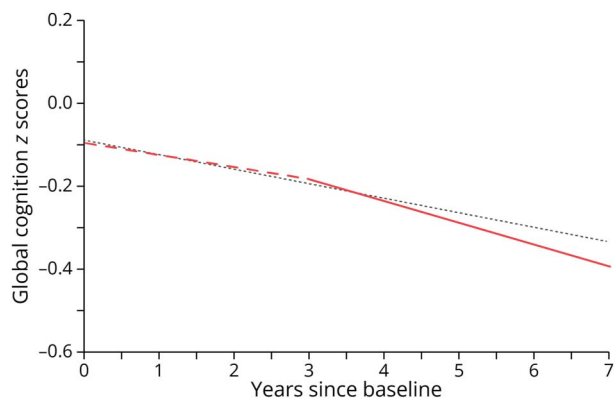
Information on sociodemographic factors, lifestyle behaviors, and health-related variables was collected. Our analysis adjusted baseline covariates associated with diabetes or cognitive

function, including age, sex, education, marital status, residential area, smoking, drinking, the number of instrumental activities of daily living (IADLs), hypertension, high total cholesterol, lung diseases, heart problems, cancer, and depressive symptoms. Details of covariates are available in eAppendix 2 (links.lww.com/WNL/D458).¹⁷

Statistical Analysis

Baseline characteristics were compared between participants who developed diabetes during the follow-up and those free of diabetes until the end of the follow-up. We constructed a linear mixed-effect model, including both fixed effects and random effects, to explore the longitudinal data with repeated measurement.^{20,21} Time and participant are the level 1 and level 2 variables, respectively. We fitted fixed effects for intercept, diabetes (with or without during the follow-up), years since baseline, "diabetes*years since baseline" interaction, "diabetes*diabetes status*years after diabetes" interaction, and all covariates. Intercept and slope (years since baseline and years after diabetes) are fitted for both fixed effects and random effects. The variable "diabetes" was set to 0 for the control group and 1 for the diabetes group. The fixed effect parameter estimate for "diabetes" could reflect the baseline difference between the diabetes group and the control group. The "years after diabetes" was set to 0 for the control group and for the diabetes group during the prediabetes period, and it indicated whether the average cognitive slope changed after diabetes onset compared with the average slope before diabetes. After

Figure 2 Predicted Changes in Global Cognition Z-Scores During the Follow-Up

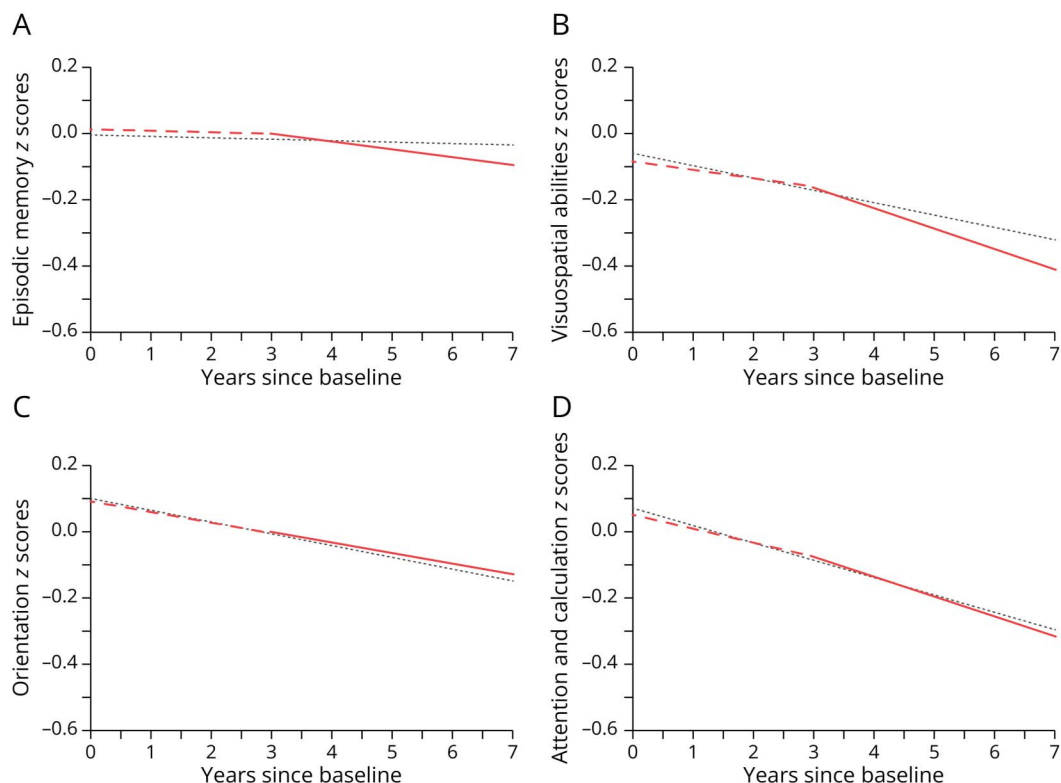


The black dotted curves represent individuals free of diabetes. The red dashed curves reflect the cognitive trends before diabetes onset among participants who developed diabetes. The red solid curves represent trends after new-onset diabetes. The predicted values of the cognitive score are calculated for a 60-year-old woman. She has finished primary school. She lives in a rural area and is married but without currently smoking, currently drinking, hypertension, dyslipidemia, lung diseases, heart problems, cancer, or depression. Her IADL score is 0. She developed diabetes at the end of the third year.

taking postdiabetes cognitive change into account, the 'diabetes*years since baseline' interaction reflected the difference in the average cognitive change rate between the diabetes group during the prediabetes period and the control group during the whole follow-up period. eFigure 1 and eAppendix 2 (links.lww.com/WNL/D458) present the conceptual model and more detailed descriptions of statistical methods. As an example, in Figures 2, 3, we show the cognitive trends of 2 fictitious persons (one with and the other without diabetes) with the most common baseline characteristics.¹⁹

To explore the association between age at diabetes onset and the extent of cognitive decline, we calculated the trends among participants who developed diabetes between 45–55, 55–65, and 65–75 years, separately. The classification was based on the sample size of our diabetes participants across ages, and a previous article reporting that compared with those with diabetes onset at age 70 years, those developing diabetes onset earlier had a higher risk of subsequent dementia.¹⁶ Using interaction terms, we tested whether the age group modified the effect of diabetes on cognitive trajectories.²² We further explore the interaction effect of the potential risk factors for cognitive decline, including sex,

Figure 3 Predicted Changes in Z-Scores of 4 Cognitive Tests During the Follow-Up



The black dotted curves represent individuals free of diabetes. The red dashed curves reflect the cognitive trends before diabetes onset among participants who developed diabetes. The red solid curves represent trends after new-onset diabetes. The predicted values of the cognitive score are calculated for a 60-year-old woman. She has finished primary school. She lives in a rural area and is married but without currently smoking, currently drinking, hypertension, dyslipidemia, lung diseases, heart problems, cancer, or depression. Her IADL score is 0. She developed diabetes at the end of the third year.

cognitive reserve (education and baseline cognitive scores), residential areas, and medication after diabetes (eFigure 2, links.lww.com/WNL/D458).

Sensitivity Analysis

For tests of global cognition, episodic memory, and attention and calculation, the cognitive decline did not follow a linear trend (eFigure 3, links.lww.com/WNL/D458, data not shown). The nonlinearity could be generated by practice effect at prior waves, which made the prediabetes rate of change of the diabetes group look more positive than it actually was. eFigure 4 shows 4 possible cognitive patterns. In eFigure 4C, if simply compared the cognitive slope in later times with the slope in early times among the diabetes group, we would achieve false-positive results that diabetes was correlated with steeper cognitive decline. Sensitivity analysis is needed to determine whether the case belonged to eFigure 4C or D. In all cognitive tests, scores of the diabetes group and the control group were similar at baseline (reported by the baseline comparison table and the linear mixed model). Therefore, as a sensitivity analysis for prediabetes difference, we compared the prediabetes cognitive slope with the slope of the control group from 0 to 2 years, on the ground of the average prediabetes follow-up time. Moreover, we verified whether the score change between the first and the last time point (wave 1 and wave 4) differed between the diabetes group and the control group as a sensitivity analysis for postdiabetes cognitive change.

The diabetes group had longer follow-ups because they were “about to be diagnosed with diabetes.” All diabetes participants were alive before the time of diabetes onset, but participants in the control group might lose to follow-up or have already died at the same time. Those lost to follow-up had poorer cognitive scores (eTable 1, links.lww.com/WNL/D458), probably making us underestimate the cognitive scores of the control group before diabetes. Therefore, we conducted a match analysis in which participants who developed diabetes (case) during follow-up were matched with 3 randomly selected participants from the control group without diabetes (control) who were still alive in the earliest wave after the diabetes onset. Matching was based on sex and same age (± 1 year) of cases and their matched controls at the diabetes time. To evaluate the stability of our primary results, the main analysis was repeated separately using the following procedures: (1) Adding “diabetes*diabetes status” as a fixed effect in the linear mixed model. The “diabetes status” is a time-varying dichotomous variable that changed from “0” to “1” at the time of diabetes onset and reflected whether the cognitive function declined acutely at diabetes onset.^{23,24} A detailed description of this variable is presented in eFigure 5 and eAppendix 2.²⁵ (2) Further adjusting for body mass index (BMI). Since few participants had BMI data in wave 1, measures of BMI in wave 2 were used as the estimates in wave 1 in this analysis. (3) Excluding participants who had a stroke during the follow-up. (4) Restricting participants to those who had complete cognitive data in all 4 waves.

Table 1 Number of Diabetes Onset According to Different Definitions

Diagnosis time	Number of new-onset diabetes
Self-reported (+) ^a , y	
0–2	259
2–4	106
4–7	357
Self-reported (–) Medication (+) ^b , y	
0–2	0
2–4	68
4–7	64
Self-reported (–) Medication (–) HbA1c (+), y	
0	64 ^c
0–4	353

^a Reported diagnosis time of diabetes by a doctor and the diagnosis year. 6 of these participants did not report diabetes diagnosis in one wave but reported diabetes in the next wave.
^b Did not report diagnosis of diabetes or medication use in one wave but reported use of insulin or antihyperglycemic medication in the next wave.
^c Did not report diagnosis of diabetes, insulin usage, or antihyperglycemic medication. But HbA_{1c} $\geq 6.5\%$. 64 participants were excluded from this study due to baseline diabetes.

All statistical analyses were performed using SAS version 9.4 (SAS Institute Inc., Cary, NC). Two-sided *p* values below 0.05 were considered statistically significant.

Data Availability

The data for CHARLS are available through a procedure at: charls.pku.edu.cn.

Results

Study Participants

Among the 12,422 participants included at baseline, 1,207 (9.7%) developed new-onset diabetes during the follow-up. The number and diagnosis time according to definitions of diabetes are presented in Table 1. 716 participants reported their diagnosis year, and 491 participants’ diagnosis time was estimated by the midpoint. The mean $\pm$ age of individuals developing diabetes was 59.1 $\pm$ 8.6 years, and 39.9% of them were male participants (Table 2).

From waves 1 to 4, the available cognition measurements were 11,215, 9,974, 9,675, and 7,678 in the without diabetes group and 1,207, 1,112, 1,120, and 930 in the diabetes group, respectively (eTable 2, links.lww.com/WNL/D458). The median follow-up time was 7.0 years (interquartile range: 7 to 7 years) for the diabetes group and 7.0 years (interquartile range: 4–7 years) for the control group. Diabetes developed at a mean of 3.2 years (SD, 2.0 years) after baseline. The mean (SD) time

Table 2 Baseline Participant Characteristics According to Diabetes Onset

	Control group (n = 11,215)	Diabetes group (n = 1,207)	p Value ^a	Adjusted p value ^b
Continuous variables, mean (SD)				
Age	58.4 (9.3)	59.1 (8.6)	0.016	None
Number of IADLs	0.1 (0.6)	0.2 (0.6)	0.024	0.091
Cognitive test scores				
Episodic memory	3.3 (1.9)	3.2 (1.9)	0.232	0.816
Visuospatial abilities	0.7 (0.5)	0.6 (0.5)	0.181	0.735
Orientation	3.8 (1.4)	3.7 (1.4)	0.008	0.201
Attention and calculation	2.8 (2.0)	2.7(2.0)	0.050	0.826
Categorical variables, n (%)				
Male participants	5,413 (48.3)	481 (39.9)	<0.001	None
Education			0.004	0.351
Illiterate	2,943 (26.3)	367 (30.4)		
Primary school	4,517 (40.3)	481 (40.0)		
Middle school	2,662 (21.1)	239 (19.8)		
High school and above	1,386 (12.4)	120 (9.9)		
Marital status			0.013	0.945
Married	9,961 (88.8)	1,062 (88.0)		
Other status	1,254 (11.3)	145 (12.0)		
Residential area			0.524	0.439
Urban	2,280 (20.4)	263 (19.6)		
Rural	8,918 (79.6)	969 (80.4)		
Current smoking	4,444 (39.6)	419 (34.7)	<0.001	0.579
Current drinking	2,952 (26.3)	263 (21.8)	<0.001	0.300
Hypertension	2,991 (26.7)	492(40.7)	<0.001	<0.001
High total cholesterol	1,462 (13.0)	271 (22.5)	<0.001	<0.001
Cancer	87 (0.8)	16 (1.3)	0.019	0.089
Lung diseases	1,067 (9.5)	133 (11.0)	0.093	0.060
Heart problems	1,118 (10.0)	193 (16.0)	<0.001	<0.001
Depression	2,268 (20.2)	298 (24.7)	<0.001	0.006

^a Calculated using analysis of variance, χ^2 , or Fisher test.
^b Calculated by generalized linear models or logistic regression after adjusting for baseline age and sex.

from the baseline cognitive assessment to the last prediabetes assessment was 2.1 (1.5) years, while the mean (SD) time between the last prediabetes assessment and diabetes onset was 1.1 (1.0) years. The diabetes group received a mean number of 1.6 (SD, 0.8) cognitive tests after diabetes onset.

Trajectories of Cognitive Scores

For those without diabetes, the cognitive scores declined annually, which is shown by the black dotted lines in Figure 2.

No difference in baseline scores was observed between the diabetes group and the control group. The decline rate from baseline to diabetes onset of the diabetes group was not statistically different from that of the control group from baseline to the end of follow-up (Table 3, eTable 3, [links.lww.com/WNL/D458](https://www.lww.com/WNL/D458)). Hence, the cognitive scores of the diabetes group also declined annually before diabetes. The above results illustrated that there was no prediabetes cognitive disadvantage in our diabetes group.

Table 3 Difference in Cognitive Z-Scores Over Time Between the Diabetes Group and the Nondiabetes Group^a

	Baseline difference		Difference in slope before diabetes onset		Difference in slope after diabetes onset	
	β (95% CI)	<i>p</i> Value	β (95% CI)	<i>p</i> Value	β (95% CI)	<i>p</i> Value
Global cognition	−0.006 (−0.055 to 0.042)	0.802	0.006 (−0.007 to 0.019)	0.376	−0.023 (−0.043 to −0.004)	0.019
Episodic memory	0.018 (−0.039 to 0.076)	0.530	0.000 (−0.016 to 0.015)	0.951	−0.018 (−0.041 to 0.004)	0.116
Visuospatial abilities	−0.025 (−0.082 to 0.033)	0.400	0.011 (−0.005 to 0.028)	0.178	−0.036 (−0.061 to −0.011)	0.004
Orientation	−0.009 (−0.057 to 0.039)	0.713	0.003 (−0.009 to 0.016)	0.603	0.001 (−0.018 to 0.020)	0.894
Attention and calculation	−0.019 (−0.074 to 0.036)	0.496	0.010 (−0.004 to 0.024)	0.174	−0.017 (−0.037 to 0.003)	0.090

^a Adjusted for baseline age, sex, education, marital status, residential area, current smoking, current drinking, hypertension, dyslipidemia, lung diseases, heart problems, cancer, depression, and number of IADLs.

In several years after diabetes onset, the cognitive scores declined over time (Figure 2). For the control group, the global cognitive scores declined by a mean of 0.035 SD/year (95% CI −0.038 to −0.032; $p < 0.001$; eTable 3, [links.lww.com/WNL/D458](https://www.lww.com/WNL/D458)). The slope of global cognition scores after diabetes onset was steeper than it was before diabetes (β , −0.023 SD/year; 95% CI −0.043 to −0.004; $p = 0.019$; Table 3 and Figure 2), roughly 68% faster (95% CI 44%–90%). Scores of visuospatial abilities test also declined faster after diabetes (β , −0.036 SD/year; 95% CI −0.061 to −0.011; $p = 0.004$), about 96% faster (95% CI 68%–125%). Compared with the pre-diabetes slope, participants who developed diabetes did not

demonstrate statistically significant faster declines in tests of episodic memory (β , −0.018 SD/year; 95% CI −0.041 to 0.004; $p = 0.116$), attention and calculation (β , −0.017 SD/year; 95% CI −0.037 to 0.003; $p = 0.090$), or orientation (β , 0.001 SD/year; 95% CI −0.018 to 0.020; $p = 0.894$).

Risk Factors for Cognitive Decline

There were 275 participants developed diabetes between 45–54 years, and 478 participants between 55–64 years, and 352 participants between 65–74 years, respectively. Changes in cognitive slope after diabetes for the 3 age groups in global cognition scores were −0.028 SD/year (95% CI −0.068 to

Table 4 Changes in Cognitive Z-Scores After Diabetes According to Age at Diabetes Onset^a

	45–54 y (n = 275)	55–64 y (n = 478)	65–74 y (n = 352)
Global cognition			
β (95% CI)	−0.028 (−0.068 to 0.011)	−0.031 (−0.062 to −0.001)	−0.030 (−0.067 to 0.008)
<i>p</i> for interaction ^b	Ref.	0.701	0.996
Episodic memory			
β (95% CI)	−0.014 (−0.063 to 0.034)	−0.049 (−0.085 to −0.013)	−0.021 (−0.064 to 0.021)
<i>p</i> for interaction	Ref.	0.576	0.903
Visuospatial abilities			
β (95% CI)	−0.037 (−0.086 to 0.013)	−0.046 (−0.085 to −0.007)	−0.028 (−0.076 to 0.019)
<i>p</i> for interaction	Ref.	0.651	0.491
Orientation			
β (95% CI)	0.007 (−0.034 to 0.048)	0.019 (−0.012 to 0.05)	−0.018 (−0.055 to 0.02)
<i>p</i> for interaction	Ref.	0.614	0.211
Attention and calculation			
β (95% CI)	−0.029 (−0.079 to 0.020)	−0.015 (−0.048 to 0.018)	−0.015 (−0.052 to 0.022)
<i>p</i> for interaction	Ref.	0.170	0.460

^a Adjusted for baseline age, sex, education, marital status, residential area, current smoking, current drinking, hypertension, dyslipidemia, lung diseases, heart problems, cancer, depression, and number of IADLs.

^b The interaction term reflected whether age groups modified the effect of diabetes on the cognitive decline rate after diabetes onset.

0.011; $p = 0.104$), -0.031 SD/year (95% CI -0.062 to -0.001 ; $p = 0.041$), and -0.030 SD/year (95% CI -0.067 to 0.008 ; $p = 0.124$), respectively. Compared with participants who developed diabetes between 45–54 years, those developing diabetes between 55–64 years (p for interaction = 0.701) or between 65–74 years (p for interaction = 0.996) did not experience significant increment in cognitive decline in the years following diabetes onset (Table 4).

Female participants experienced an accelerated decline in global cognition after diabetes (β , -0.035 SD/year; 95% CI -0.065 to -0.006 ; $p = 0.019$), whereas male participants did not show significant accelerated decline (β , -0.016 SD/year; 95% CI -0.041 to -0.010 ; $p = 0.203$). Nevertheless, there was no significant difference between male and female participants (p for interaction = 0.288). Residential area did not modify the effect of diabetes on postdiabetes cognitive change (p for interaction = 0.820). Compared with participants who had not finished primary school, those who finished middle school (p for interaction = 0.012) or high school (p for interaction <0.001) showed slower after-diabetes-cognitive-decline in episodic memory scores, but not in global cognition or other tests. Participants in higher baseline cognitive score tertiles had slower cognitive declines after diabetes, compared with participants in the lowest cognitive score tertile (p for interaction <0.001). Individuals using antidiabetic agents experienced a faster decline in global cognition after diabetes (β , -0.044 SD/year; 95% CI -0.075 to -0.013 ; $p = 0.003$), while patients not using antidiabetic agents did not show postdiabetes cognitive deterioration (β , -0.007 SD/year; 95% CI -0.031 to 0.018 ; $p = 0.578$). Those using antidiabetic agents disclosed steeper decline (p for interaction = 0.037, eTables 4–8, [links.lww.com/WNL/D458](https://www.lww.com/WNL/D458)).

Sensitivity Analyses

When comparing the slope between 2 time points (0-year/wave 1 and 2-year/wave 2) before diabetes, the average prediabetes cognitive slope of the diabetes group was similar to that of the control group (eTable 9, [links.lww.com/WNL/D458](https://www.lww.com/WNL/D458)). We then compared the slope between wave 1 and wave 4. The mean global cognitive slope of the diabetes group was less than that of the control group (β , -0.049 SD; 95% CI -0.085 to -0.013 ; $p = 0.008$; eTable 10). This is in accordance with our primary result that global cognition deteriorated after new-onset diabetes. Similar results were found in tests of episodic memory (a tendency to deteriorate after diabetes), visuospatial abilities (deteriorated after diabetes), and orientation (did not change after diabetes). However, for the attention and calculation test, the slope between wave 1 and wave 4 was similar between the 2 groups (β , 0.003 SD; 95% CI -0.033 to 0.040 ; $p = 0.848$; eTable 10), suggesting that its decline tendency was ‘false-positive.’

In the match analysis, the diabetes group did not exhibit prediabetes cognitive disadvantage compared with the match group without diabetes. However, the postdiabetes cognitive decline of global cognition score turned out to be

statistically insignificant (β , -0.021 SD/year; 95% CI -0.042 to 0.001 ; $p = 0.059$; eTable 11, [links.lww.com/WNL/D458](https://www.lww.com/WNL/D458)). We did not observe “acute cognitive decline/change” at the time of diabetes onset (global cognition; β , 0.007 SD/year; 95% CI -0.055 to 0.069 ; $p = 0.828$, eTable 12). After adding BMI as a covariate, excluding participants who had a stroke during the follow-up, or restricting participants to those receiving cognitive measurements in all 4 waves, we achieved constant results that scores of global cognition and visuospatial abilities declined gradually before diabetes and showed a faster deterioration rate in the years after diabetes onset (eTables 13–15).

Discussion

In this population-based study of Chinese adults aged 45 years or older, participants experienced faster rates of decline in global cognition and visuospatial ability scores several years after diabetes onset, compared with the rate before diabetes. However, the postdiabetes cognitive decline was not significantly steeper in tests of episodic memory, attention and calculation, or orientation. The rate of decline before diabetes onset among those with diabetes was similar to the rate of individuals free of diabetes.

The cognitive trajectory before and after new-onset diabetes was relatively unexplored, especially in Asia. Several longitudinal studies have assessed their participants’ cognitive function more than once and calculated the cognitive trends of those who developed new-onset diabetes during the follow-up period.^{9–13} However, they rarely analyze whether the trends in cognition after diabetes onset differed from trends before diabetes. To our knowledge, only 2 studies compared the cognitive trajectory before and after diabetes onset. A Chicago Health and Aging Project (CHAP) study reported faster postdiabetes declines in Mini-Mental State Examination (MMSE) scores, executive function, and episodic memory.¹⁴ A recent study using data from the English Longitudinal Study of Ageing (ELSA) suggested accelerated declines in tests of global cognition, orientation, memory, and executive function after diabetes onset. A key finding of our study lies in the different patterns of cognitive decline. Different from the ELSA study from the United Kingdom, the orientation ability scores in our study did not deteriorate after diabetes.¹⁵ We were unable to explore the difference. Future neuroimaging studies could learn the reasons for different cognitive tests. In addition, this study reported the slope of visuospatial abilities steeper after diabetes, providing a fuller picture of the cognitive patterns. Our results of the episodic memory and attention and calculation did not meet our threshold of significance, which might be owing to different disease duration and the length of follow-up. In accordance with the previous 2 studies, accelerated cognitive decline occurs after but not before diabetes onset. Close monitoring of cognitive dysfunction in patients with diabetes is vitally important in short periods after a diabetes diagnosis.

Another important finding was our subgroup analyses by age. A Whitehall II study linked every 5-year younger age at diabetes onset to a higher risk of dementia at age 70 years (hazard ratio, 1.24; 95% CI 1.06–1.46).¹⁶ This intrigued us to investigate whether younger diabetes patients had increased cognitive decline compared with older patients. However, we did not find age groups modifying the effect of diabetes onset on the changes in cognitive slope after diabetes. The previous Whitehall II study collected data on diabetes onset and dementia diagnosis from 1985 to 2019. The follow-up period of CHARLS is relatively shorter. Our study could be updated after CHARLS releases its new survey in the future and explore whether new findings could be achieved after the timeline is prolonged.

Although the exact pathophysiologic pathway through which diabetes contributes to cognitive decline beyond aging remains unclear, several mechanisms have been proposed.²⁶ Increasing evidence suggests that insulin resistance might be a key factor. Brain insulin resistance may lead to impairments in synaptic, metabolic, and immune response functions, which further damage cognitive function.²⁷ Moreover, diabetes induces Alzheimer disease–related pathology, such as β -amyloid neurotic plaques (NPs) and intracellular neurofibrillary tangles (NFTs).²⁸ Another vital mechanism linking hyperglycemia to cognitive impairment is various vascular damages such as stroke, large vessel arteriosclerosis, small vessel arteriosclerosis, and cerebral amyloid angiopathy. Studies also reported that increased white matter hyperintensities and lacunar infarcts were observed in diabetes patients, which were associated with cognitive impairment.^{26,27,29,30} In addition, oxidative stress, neuroinflammation, and some risk factors accompanying diabetes such as hypertension, obesity, and depression all lead to cognition impairment.^{26,31–34} In summary, multiple mechanisms jointly contribute to brain injury, eventually resulting in postdiabetes cognitive decline. Previous stroke studies reported that stroke patients experienced an acute decline at stroke onset but not accelerated future decline in global cognition.^{23,24} By contrast, diabetes patients did not exhibit “acute cognitive decline,” illustrating that the adverse impact of diabetes-related pathology on the brain tends to be a gradual process, instead of a sudden brain injury such as stroke.

This study has several advantages. It is one of the largest studies exploring the cognitive trends during the prediabetes period and in the years after new-onset diabetes. The large sample size with repeated measurements offered us the opportunity to generate precise and reliable trajectories of cognitive function. Furthermore, our results extend the findings of previous studies by subgroup analyses of younger participants.

A few limitations of this study should be mentioned. First, the follow-up period of CHARLS is relatively short. Diabetes is a life-long condition. Our results could only reflect the cognitive trends in the short term after diabetes. The cognitive

assessments are limited, and diabetes diagnoses for part of the participants are imprecise. Many of our participants were from rural areas. The generalizability to other populations or the long term after diabetes might be a concern. Second, for part of cognitive tests, relationships between time and scores were nonlinear. Accordingly, the linear mixed model could only illustrate that participants experienced a slight cognitive deterioration after developing diabetes but could not visualize the actual cognitive trends. Because of limited follow-up times, we were unable to delete or adjust for baseline scores to mitigate the practice effects which led to a boost in performance after the first test.³⁵ Third, from wave 1 to wave 3, the CHARLS lack tests such as MMSE, which precluded us from defining dementia or mild cognitive impairment (MCI) by using cut-off points of cognitive scores.^{36,37} Among our participants, the cognitive trajectories were similar across different age subgroups. However, we were unable to study whether younger participants had higher risk of dementia, which is a topical research area. Fourth, CHARLS only performed blood tests in wave 1 and wave 3. A certain proportion of diabetes patients were unaware of their glucose level, leading to a delay in diagnosis of diabetes. Future studies could conduct fasting blood tests more frequently. Fifth, although a variety of confounding factors were adjusted, there may still be some related but uncontrolled variables such as apolipoprotein E (ApoE) genotype and air pollution. The frequency of the ApoE ϵ 4 allele is approximately 7% in the Chinese population.³⁸ Consequently, the genotype may have fewer effects on our results.

In conclusion, our findings demonstrated that new-onset diabetes was associated with a faster rate of future cognitive decline, but not during the prediabetes period. Efforts in eliminating the adverse impacts on cognition should be started on diagnosis of diabetes. Future studies are needed to learn the mechanisms of cognitive decline in a few years after new-onset diabetes.

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References

- Patterson C. *Alzheimer's Disease International*. Updated London: 2018. Accessed May 21, 2023. apo.org.au/node/260056.
- Livingston G, Huntley J, Sommerlad A, et al. Dementia prevention, intervention, and care: 2020 report of the Lancet Commission. *Lancet*. 2020;396(10248):413-446. doi: 10.1016/S0140-6736(20)30367-6
- Scheltens P, Blennow K, Breteler MM, et al. Alzheimer's disease. *Lancet*. 2016; 388(10043):505-517. doi:10.1016/S0140-6736(15)01124-1
- O'Brien JT, Thomas A. Vascular dementia. *Lancet*. 2015;386(10004):1698-1706. doi: 10.1016/S0140-6736(15)00463-8
- Xue M, Xu W, Ou YN, et al. Diabetes mellitus and risks of cognitive impairment and dementia: a systematic review and meta-analysis of 144 prospective studies. *Ageing Res Rev*. 2019;55:55100944. doi:10.1016/j.arr.2019.100944
- Ganmore I, Beeri M. *Magnitude and Trajectories of Cognitive Dysfunction in Type 2 Diabetes Mellitus*. Updated 2018. Accessed June 1, 2023. doi.org/10.1016/B978-0-12-809454-9.00003-2.
- Fontbonne A, Berr C, Ducimetiere P, Alperovitch A. Changes in cognitive abilities over a 4-year period are unfavorably affected in elderly diabetic subjects: results of the Epidemiology of Vascular Aging Study. *Diabetes Care*. 2001;24(2):366-370. doi: 10.2337/diacare.24.2.366
- Hu J, Fang M, Pike JR, et al. Prediabetes, intervening diabetes and subsequent risk of dementia: the Atherosclerosis Risk in Communities (ARIC) study. *Diabetologia*. 2023;66(8):1442-1449. doi:10.1007/s00125-023-05930-7
- Rawlings AM, Sharrett AR, Schneider AL, et al. Diabetes in midlife and cognitive change over 20 years: a cohort study. *Ann Intern Med*. 2014;161(11):785-793. doi: 10.7326/M14-0737
- Spauwen PJ, Kohler S, Verhey FR, Stehouwer CD, van Bostel MP. Effects of type 2 diabetes on 12-year cognitive change: results from the Maastricht Aging Study. *Diabetes Care*. 2013;36(6):1554-1561. doi:10.2337/dcl2-0746
- Mayeda ER, Haan MN, Neuhaus J, et al. Type 2 diabetes and cognitive decline over 14 years in middle-aged African Americans and Whites: the ARIC Brain MRI Study. *Neuroepidemiology*. 2014;43(3-4):220-227. doi:10.1159/000366506
- Yaffe K, Falvey C, Hamilton N, et al. Diabetes, glucose control, and 9-year cognitive decline among older adults without dementia. *Arch Neurol*. 2012;69(9):1170-1175. doi:10.1001/archneurol.2012.1117
- Mayeda ER, Haan MN, Yaffe K, Kanaya AM, Neuhaus J. Does type 2 diabetes increase rate of cognitive decline in older Mexican Americans? *Alzheimer Dis Assoc Disord*. 2015;29(3):206-212. doi:10.1097/WAD.0000000000000083
- Rajan KB, Arvanitakis Z, Lynch EB, et al. Cognitive decline following incident and preexisting diabetes mellitus in a population sample. *Neurology*. 2016;87(16):1681-1687. doi:10.1212/WNL.0000000000003226
- Ji X, Gao H, Sun D, et al. Trajectories of cognition and daily functioning before and after incident diabetes. *Diabetes Care*. 2023;46(1):75-82. doi:10.2337/dc22-1190
- Barbiellini Amidei C, Fayosse A, Dumurgier J, et al. Association between age at diabetes onset and subsequent risk of dementia. *JAMA*. 2021;325(16):1640-1649. doi:10.1001/jama.2021.4001
- Zhao Y, Hu Y, Smith JP, Strauss J, Yang G. Cohort profile: the China health and Retirement longitudinal study (CHARLS). *Int J Epidemiol*. 2014;43(1):61-68. doi: 10.1093/ije/dys203
- Hua J, Qiao Y, Ke C, Shen Y. Higher visit-to-visit total cholesterol variability is associated with lower cognitive function among middle-aged and elderly Chinese men. *Sci Rep*. 2020;10(1):15555. doi:10.1038/s41598-020-72601-7
- Zheng F, Yan L, Zhong B, Yang Z, Xie W. Progression of cognitive decline before and after incident stroke. *Neurology*. 2019;93(1):e20-e28. doi:10.1212/wnl.00000000000007716
- West BT, Welch KB, Galecki AT. *Linear Mixed Models: A Practical Guide Using Statistical Software*. Updated 2007. Accessed August 1, 2021. doi.org/10.1201/9781003181064.
- Laird NM, Ware JH. Random-effects models for longitudinal data. Updated 1982. Accessed December 1, 2021. ncbi.nlm.nih.gov/pubmed/7168798.
- Levine DA, Wadley VG, Langa KM, et al. Risk factors for poststroke cognitive decline: the REGARDS study (reasons for geographic and racial differences in stroke). *Stroke*. 2018;49(4):987-994. doi:10.1161/STROKEAHA.117.018529
- Levine DA, Kabeto M, Langa KM, Lisabeth LD, Rogers MA, Galecki AT. Does stroke contribute to racial differences in cognitive decline? *Stroke*. 2015;46(7):1897-1902. doi:10.1161/STROKEAHA.114.008156
- Hua J, Dong J, Chen GC, Shen Y. Trends in cognitive function before and after stroke in China. *BMC Med*. 2023;21(1):204. doi:10.1186/s12916-023-02908-5
- Pendlebury ST, Rothwell PM. Prevalence, incidence, and factors associated with pre-stroke and post-stroke dementia: a systematic review and meta-analysis. *Lancet Neurol*. 2009;8(11):1006-1018. doi:10.1016/S1474-4422(09)70236-4
- Biessels GJ, Despa F. Cognitive decline and dementia in diabetes mellitus: mechanisms and clinical implications. *Nat Rev Endocrinol*. 2018;14(10):591-604. doi: 10.1038/s41574-018-0048-7
- Arnold SE, Arvanitakis Z, Macauley-Rambach SL, et al. Brain insulin resistance in type 2 diabetes and Alzheimer disease: concepts and conundrums. *Nat Rev Neurol*. 2018; 14(3):168-181. doi:10.1038/nrneuro.2017.185
- Umegaki H. Type 2 diabetes as a risk factor for cognitive impairment: current insights. *Clin Interv Aging*. 2014;9:1011-1019. doi:10.2147/CIA.S48926
- Abner EL, Nelson PT, Kryscio RJ, et al. Diabetes is associated with cerebrovascular but not Alzheimer's disease neuropathology. *Alzheimers Dement*. 2016;12(8):882-889. doi:10.1016/j.jalz.2015.12.006
- van Sloten TT, Sedaghat S, Carnethon MR, Launer LJ, Stehouwer CDA. Cerebral microvascular complications of type 2 diabetes: stroke, cognitive dysfunction, and depression. *Lancet Diabetes Endocrinol*. 2020;8(4):325-336. doi:10.1016/S2213-8587(19)30405-X
- Biessels GJ, Strachan MW, Visseren FL, Kappelle LJ, Whitmer RA. Dementia and cognitive decline in type 2 diabetes and prediabetic stages: towards targeted interventions. *Lancet Diabetes Endocrinol*. 2014;2(3):246-255. doi:10.1016/S2213-8587(13)70088-3
- Dutta BJ, Singh S, Seksaria S, Das Gupta G, Singh A. Inside the diabetic brain: insulin resistance and molecular mechanism associated with cognitive impairment and its possible therapeutic strategies. *Pharmacol Res*. 2022;182:182106358. doi:10.1016/j.phrs.2022.106358
- Ganguli M, Beer JC, Zmuda JM, et al. Aging, diabetes, obesity, and cognitive decline: a population-based study. *J Am Geriatr Soc*. 2020;68(5):991-998. doi:10.1111/jgs.16321
- Srikanth V, Sinclair AJ, Hill-Briggs F, Moran C, Biessels GJ. Type 2 diabetes and cognitive dysfunction-towards effective management of both comorbidities. *Lancet Diabetes Endocrinol*. 2020;8(6):535-545. doi:10.1016/S2213-8587(20)30118-2
- Faul JD, Ware EB, Kabeto MU, Fisher J, Langa KM. The effect of childhood socioeconomic position and social mobility on cognitive function and change among older adults: a comparison between the United States and England. *J Gerontol B Psychol Sci Soc Sci*. 2021;76(suppl 1):S51-S63. doi:10.1093/geronb/gbaa138
- Folstein MF, Folstein SE, McHugh PR. "Mini-mental state". A practical method for grading the cognitive state of patients for the clinician. *J Psychiatr Res*. 1975;12(3): 189-198. doi:10.1016/0022-3956(75)90026-6
- Jia X, Wang Z, Huang F, et al. A comparison of the Mini-Mental State Examination (MMSE) with the Montreal Cognitive Assessment (MoCA) for mild cognitive impairment screening in Chinese middle-aged and older population: a cross-sectional study. *BMC Psychiatry*. 2021;21(1):485. doi:10.1186/s12888-021-03495-6
- Ji Y, Liu M, Huo YR, et al. Apolipoprotein E ε4 frequency is increased among Chinese patients with frontotemporal dementia and Alzheimer's disease. *Dement Geriatr Cogn Disord*. 2013;36(3-4):163-170. doi:10.1159/000350872